

Q. Comparative analysis between Batch & Streamed data processing tools like Map-reduce, Apache spark, Apache Flink, Apache Samza, Apache Kafka & Apache storm.

→ Data processing can be categorized into batch processing & stream processing:

- i) Batch Processing: Processes large volumes of data in fixed time intervals
- ii) Stream Processing: Processes data continuously in real-time

I) Batch Processing Tools

	Feature	Map-reduce	Apache Spark
i)	Processing Type	Batch Processing	Batch & Stream Processing
ii)	Speed	Slower due to multiple disk read/writes	Faster due to in-memory processing
iii)	Fault Tolerance	Checkpoint-based recovery	DAG-based lineage recovery
iv)	Ease of Use	Complex requires Java	Easier supports multiple language like Python, Scala
v)	Use case	Large-scale data analytics (e.g. log analysis)	Machine learning, real-time analytics

- ① Batch processing is useful for large-scale offline data analysis.
- ② Apache Spark is the most versatile, supporting both batch & stream processing.

II) Stream Processing Tools

	Feature	Apache Flink	Apache Samza	Apache Kafka	Apache Storm
i)	Processing Type	Stream & Batch	Stream	Stream	Stream
ii)	Speed	Low latency, high throughput	High throughput	High throughput	Low latency real-time
iii)	Fault Tolerance	Exactly-once processing	Checkpointing with Kafka	Replication & durability	Tuple-level ACKing
iv)	Ease of Use	Complex but powerful	Requires Kafka integration	Easy to use	Difficult to configure
v)	Use Case	Real-time analytics, event-driven applications	Log processing, messaging systems	Message queuing, data pipeline	Realtime monitoring, fraud detection

① Stream processing is needed for real-time event processing

② Kafka + Flink is a powerful combination for real-time data pipelines

Each tool is designed for specific use cases & the choice depends on latency requirements, fault tolerance & ease of integration

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